



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid Spring Semester Examination 2023-24

Date of Examination: 21/02/2024 Session: (FN/AN) AN Duration: 2 Hrs Full Marks: 40
Subject No. : A161002 Subject : Deep Learning Foundations and Applications
Department/Center/School: Center of Excellence in Artificial Intelligence
Specific charts, graph paper, log book etc., required No
Special Instructions (if any) : _____

Instructions

1. Except electronic calculators, no other devices or materials are allowed during the exam.
2. All answers must be supported by calculations and/or explanations unless stated otherwise.

Question 1:

Pick the correct option(s) in each of the following questions. Note that there may be more than one correct option associated with each question, and in such cases you must specify all the options to receive full credit. Explanations are not necessary for this question.

1. ☒ The resurgence of deep learning could be attributed to the following factors [1 point]
 - i) improved hardware capabilities
 - ii) improved computational models of the human brain
 - iii) abundance of training data
 - iv) large scale investments
2. ☒ Which of the following are effective strategies to mitigate overfitting? [1 point]
 - i) Data augmentation
 - ii) L_2 regularization
 - iii) increasing model capacity
 - iv) All of the above
3. ☒ Consider a fully-connected neural network with a single hidden layer and an output layer. Each hidden unit employs the logistic activation function, and all weights and biases are initialized to 0. The output of the network for an input $x \in \mathbb{R}^n$ would be: [1 point]
 - i) 1
 - ii) -1
 - iii) 0
 - iv) 0.5
4. ☒ Suppose that the weights of a deep feedforward neural network are initialized with very large values. Which of the following issues you might see while training the network with different activation functions? [1 point]

- i) sigmoid activation might give rise to exploding gradients, while ReLU activation might give rise to vanishing gradients.
- ii) sigmoid activation might give rise to vanishing gradient while ReLU activation is likely to work well without an issue.
- iii) sigmoid activation might give rise to vanishing gradients, while ReLU activation might give rise to exploding gradients.
- iv) Both sigmoid and ReLU activations would work well without any issues.

5. Which of the following statements are correct about KL divergence? [1 point]

- i) it can be expressed as the difference between the cross entropy of the true and estimated distributions and the entropy of the true distribution
- ii) it is symmetric
- iii) it is non-negative and unbounded from above.
- iv) All of the above

Question 2:

Please answer the following questions.

1. The true distribution of a random variable X is given as follows (w.p. stands for *with probability*):

$$p_X(x) = \begin{cases} 0 & \text{w.p. } 0.65 \\ 1 & \text{w.p. } 0.45 \\ 2 & \text{w.p. } 0.1 \end{cases}$$

Suppose the estimated distribution is the following

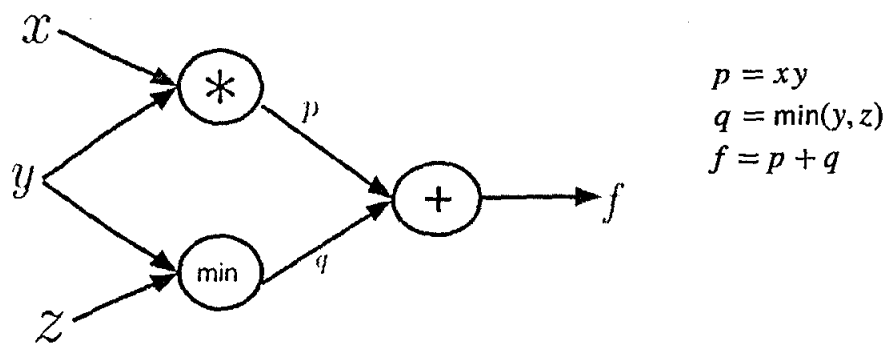
$$q_X(x) = \begin{cases} 0 & \text{w.p. } 0.3 \\ 1 & \text{w.p. } 0.5 \\ 2 & \text{w.p. } 0.2 \end{cases}$$

Compute the cross entropy $H(p, q)$. [2 points]

2. 45% of all received emails are estimated to be spam emails. A spam filter is able to successfully detect 97% of spam emails. However, 5% of non-spam emails are also detected as spam by this filter. What is the probability that an email is actually a non-spam email, given that it is detected as spam by the filter? [3 points]

3. While designing neural networks, it is often important to estimate the computational complexity of different architectures in terms of the number of floating point operations (FLOPs), where a FLOP is a single arithmetic operation like addition, subtraction, multiplication and division. Consider a fully-connected feedforward neural network having 10 input units, 20 hidden units in the first hidden layer, 15 hidden units in the second hidden layer, and 5 output units. All units have a bias, and the hidden units employ ReLU activation, while the output units employ linear activation. What is the total number of FLOPs of this architecture [2 points]

4. Consider the following computational graph and the associated functions it implements:



Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$ using backpropagation. You should re-draw the computational graph showing the relevant gradients flowing backward along the bottom of the edges in your solution.

[4 points]

Question 3: Principal Component Analysis

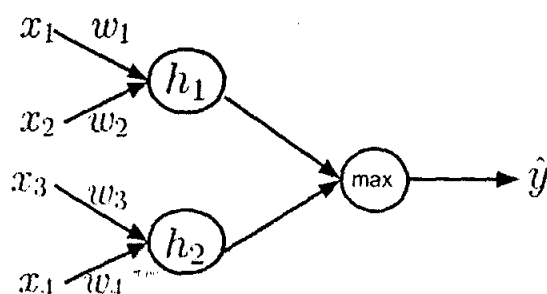
Consider a design matrix given as follows:

$$X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$$

1. Compute the centered design matrix. [1 point]
2. Calculate the unit vectors along the directions of the principal components of X . [4 points]
3. Project X along the direction of the first principal component [2 points]
4. What is the reconstruction error of the above projection? [1 point]

Question 4: Backpropagation

Consider the following neural network:



Here, h_1 and h_2 are sigmoid neurons, i.e. $h_i = \sigma(\sum w_i x_i)$ for $i = 1, 2$ and $\sigma(z) = \frac{1}{1 + e^{-z}}$. The

output $\hat{y}$ is given by $\hat{y} = \max(h_1, h_2)$. Next, consider the following inputs and weights:

$$x_1 = -0.7, x_2 = 1.2, x_3 = 1.1, x_4 = -2$$

$$w_1 = -1.7, w_2 = 0.1, w_3 = -0.6, w_4 = -1.8$$

Let the ground truth value of the prediction be $y = 0.5$ and the loss be MSE, i.e. $\mathcal{L}(y, \hat{y}) = (y - \hat{y})^2$

Using the information provided above, please answer the following questions:

1. What is the value of $\hat{y}$ on executing one forward pass of the neural network with the given values of inputs and weights? [1 points]
2. What is the value of $\frac{\partial \mathcal{L}(y, \hat{y})}{\partial \hat{y}}$? [1 point]
3. Can you express $\sigma'(z)$ (i.e. the derivative of the logistic function w.r.t its input) in terms of $\sigma(z)$? [2 points]
4. Using backpropagation, and the results you computed above, please compute the following
 - i) $\frac{\partial \mathcal{L}(y, \hat{y})}{\partial w_1}$ [1 point]
 - ii) $\frac{\partial \mathcal{L}(y, \hat{y})}{\partial w_4}$ [5 points]

[Hint: for computing the gradients using backpropagation, it would be easier to draw the computational graph of the network, followed by assigning labels to intermediate values].

Question 5: MLE for logistic regression

In class, we discussed the logistic regression model given as follows:

$$\hat{y} = \sigma(\theta^T x + b)$$

$$\text{where } \sigma(z) = \frac{1}{1 + e^{-z}}$$

Also, we assume that

$$P(y = 1 | x; \theta) = \hat{y}$$

$$P(y = 0 | x; \theta) = 1 - \hat{y}$$

Then the likelihood of a label y given an input x is given by:

$$p(y | x; \theta) = (\hat{y})^y (1 - \hat{y})^{1-y}$$

1. Assuming that there are M iid training samples given by (x^i, y^i) for $i = 1, 2, \dots, M$, write an expression for the logarithm of the likelihood function $L(\theta)$. [2 points]
2. What quantity does the above log likelihood represent? [1 point]
3. By maximizing the log likelihood, derive an expression for the update rule of the individual components of θ for logistic regression. [3 points]